

- `ballRect[0]` is the x value (but you could also use `ballRect.left`)
- `ballRect[1]` is the y value (but you could also use `ballRect.top`)
- `ballRect[2]` is the width (but you could also use `ballRect.width`)
- `ballRect[3]` is the height (but you could also use `ballRect.height`)

[Listing 5-7](#) is an alternative version of our bouncing ball program that maintains all the information about the ball in a rectangle object.

File:

PygameDemo4_OneBallBounce/PygameOneBallBounceRects.py

```
# pygame demo 4(b) - one image, bounce around the window using rects

# 1 - Import packages
import pygame
from pygame.locals import *
import sys
import random

# 2 - Define constants
BLACK = (0, 0, 0)
WINDOW_WIDTH = 640
WINDOW_HEIGHT = 480
FRAMES_PER_SECOND = 30
N_PIXELS_PER_FRAME = 3

# 3 - Initialize the world
pygame.init()
window = pygame.display.set_mode((WINDOW_WIDTH, WINDOW_HEIGHT))
clock = pygame.time.Clock()

# 4 - Load assets: image(s), sound(s), etc.
ballImage = pygame.image.load('images/ball.png')

# 5 - Initialize variables
❶ ballRect = ballImage.get_rect()
MAX_WIDTH = WINDOW_WIDTH - ballRect.width
MAX_HEIGHT = WINDOW_HEIGHT - ballRect.height
ballRect.left = random.randrange(MAX_WIDTH)
ballRect.top = random.randrange(MAX_HEIGHT)
```